

BHANU NAGPURE

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[Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio Website](#)

EDUCATION

G H Raison College of Engineering

Bachelor of Technology in Computer Science and Engineering

Nagpur, MH, India

December 2021 – June 2025

SKILLS

- **Programming:** Python, Kotlin, Java, Dart, C++, C, SQL, JavaScript, HTML, CSS
- **Frameworks:** Flutter, TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy
- **Tools:** Git, GitHub, REST APIs, Google Cloud, Android SDK, Firebase
- **Methodologies:** ML, Deep Learning, Mobile App Development, IoT, UI/UX, Agile, Project Management

PROFESSIONAL EXPERIENCE

- 1. Machine Learning Intern, Athabasca University** **Athabasca, Alberta, Canada**
Project: Ideological Analysis Algorithm August 24 - October 24
 - Engaged in machine learning research aimed at forecasting Supreme Court case outcomes using machine learning and sentiment analysis. Successfully completed the research and presented the solution to a panel of judges.
- 2. Globalink Research Intern, Mitacs** **Edmonton, Alberta, Canada**
Project: MfieldTrip - Flutter Application April 24 – August 24
 - Developed a cross-platform Flutter application for field assignments for web, iOS, and Android platforms.
 - Led end-to-end project development, including **UI/UX design, frontend and backend** implementation
 - Presented research findings at an **international conference** and written comprehensive project **documentation**
- 3. Software Developer Intern, BETIC-GHRCE** **Nagpur, MH, India**
Project: Smart device for early screening of OSMF October 22 – May 24
 - Engineered an **Android app for TMJ disorder** screening, enhancing classification accuracy by **70%** over previous systems.
 - Boosted camera **performance by 60%**, leveraging the Google ML Library to improve diagnostic accuracy
- 4. Fellowship, Indian Institute of Technology Bombay** **Mumbai, MH, India**
Project: MEDHA and MEDIC March 22 – October 22
 - Led a team to design and prototype solutions for xerostomia patients, reducing **diagnostic time by 80%** through innovative rapid prototyping techniques.
 - **Developed and presented** a functional medical device prototype using MIT App Builder, receiving commendation from industry experts at IIT Bombay.

PROJECTS

- 1. V-Safe-Anywhere: Empowering Women's Safety with Wearable AI and IoT**
Google firebase, Google sheet, Flutter, IoT, Google Apps Script November 2022 - April 2024
 - Developed an IoT-based wearable safety device, achieving real-time data sync and user authentication via Firebase, **with a 98% uptime**
 - **Published research** documenting the system's architecture, with a focus on **scalability and security**
- 2. Smart Device for Early Screening of OSMF**
Android, IoT, Firebase December 2022 - July 2023
 - Developed an Android app (OCanPredict) to support a biomedical device for OSMF screening, increasing **diagnostic efficiency by 40%**
 - Received multiple national and state-level **awards** for innovation and originality
- 3. Autonomous Floating Garbage Collection Device Using Computer Vision and Robotics**
Python, Yolo, numpy, pandas. IoT June 2023 - September 2023
 - Designed and implemented a computer vision system for detecting and retrieving floating waste, reducing **manual waste collection by 60%**.
 - Presented at **international conference**, published a paper and **open-sourced code** on GitHub

PUBLICATIONS

1. "V-Safe-Anywhere: Empowering Women's Safety with Wearable AI and IoT"

IGI Global, [Link](#)

March 1, 2024

- Explored and proposed technology-driven alternatives to traditional safety solutions, with focus on scalable and secure system

2. Dhammashree Tamgadge, Bhanu Nagpure, Riddhi Mandal, **Autonomous Floating Garbage Collection Device using Computer Vision and Robotics**. 8th International Conference on Algorithms, Computing and Systems (ICACS 2024) [Link](#)

TECHNICAL SKILLS

- **Programming:** Kotlin, Python, C, C++, Java, SQL, HTML, CSS, JavaScript
- **Frameworks & Tools:** Flutter, Firebase, Android Development, Git/GitHub
- **Development:** Cross-platform mobile development, Debugging, Testing, Documentation
- **Operating Systems:** Linux, macOS, Windows, Android

PATENTS

1. Smart Chairside TMJ Examination and Muscle Activity Detector Tool

March 28, 2023

Published, 202321022476, [Link](#)

ACHIEVEMENTS

- First Place, **TechTrek Hackathon 2025**, Nagpur January 2025
- Third Place, **Inter-University Research Convention: Avishkar 2024**, Nashik January 2024
- Runner-up at **DIPEX 2023**, Amravati April, 2023
- Runner-up at **SHRISTI 2023**, Kottayam March, 2023
- Awarded **BIREC E-YUVA GRANT** Received ₹10 lakh in funding February, 2023
- Winner, **MEDHA and MEDIC-2022** October, 2022

VOLUNTEER EXPERIENCE

Oral Cancer Awareness camps

December 2022 - June 2023

- Educated over 1,000 individuals on the risks of tobacco and betel nut consumption, collaborating with Lata Mangeshkar Hospital doctors.

CONFERENCE

3rd Annual TRESL Lab International Symposium

July 12-13, 2024

Athabasca, Alberta, Canada

- Presented research: "Development of A Mobile Field Trip Application"

8th International Conference on Algorithms, Computing and Systems

October 11-13, 2024

Hong Kong Metropolitan University, Hong Kong

- Delivered a presentation on an autonomous floating garbage collection device using computer vision and robotics, leading to increased research partnerships.

REFERENCES

- Dr. Qing Tan, Athabasca University (qingt@athabascau.ca)
- Dr. Vibha Bora, G. H. Rasoni College of Engineering (vibha.bora@raisoni.net)
- Dr. Snehlata Dongre, G. H. Rasoni College of Engineering (snehlata.dongre@raisoni.net)